

For full credit explain your reasoning, showing all relevant work.

Exercise 1. Let H denote the set of real numbers greater than 1. For $a, b \in H$ define

$$a \odot b := ab - a - b + 2$$

Show that H taken with this multiplication is a group. What is the identity element in this group? For $a \in H$ what is a^{-1} ?

Exercise 2. Let G be a group such that $a^2 = e$ for all $a \in G$ (where e is the identity element in G). Show that G is an abelian group.

Exercise 3. Let G be a group, and let $a \in G$ be an element such that $a^{12} = e$ (where e is the identity element). Find $k > 0$ such that a^k is the inverse of a^{20} . Justify your answer.

PRACTICE PROBLEMS

Exercises below are for practice only - do not turn them in for grading.

Practice Exercise 1. Let $G = \mathbb{R}^* \times \mathbb{R}$ where $\mathbb{R}$ is the set of all real numbers and $\mathbb{R}^*$ is the set of non-zero real numbers. Define a multiplication in G by the formula

$$(a, b) * (c, d) = (ac, ad + b)$$

- a) Show that G taken with this multiplication is a group. What is the identity element in this group? For $a \in H$ what is a^{-1} ?
- b) Is G an abelian group? Either prove it or give a counterexample.

Practice Exercise 2. Let G be a group with a property, that for any $a, b, c \in G$ if $ca = bc$ then $a = b$. Show that G is an abelian group.

Practice Exercise 3. Let G be a group and $a, b \in G$. Show that if $(ab)^2 = a^2b^2$ then $ab = ba$.

Practice Exercise 4. Let G be a finite group. Show that the number of elements $a \in G$ such that $a^3 = e$ is odd.